Advanced Database Project

Dr/Ahmed El-sidawy

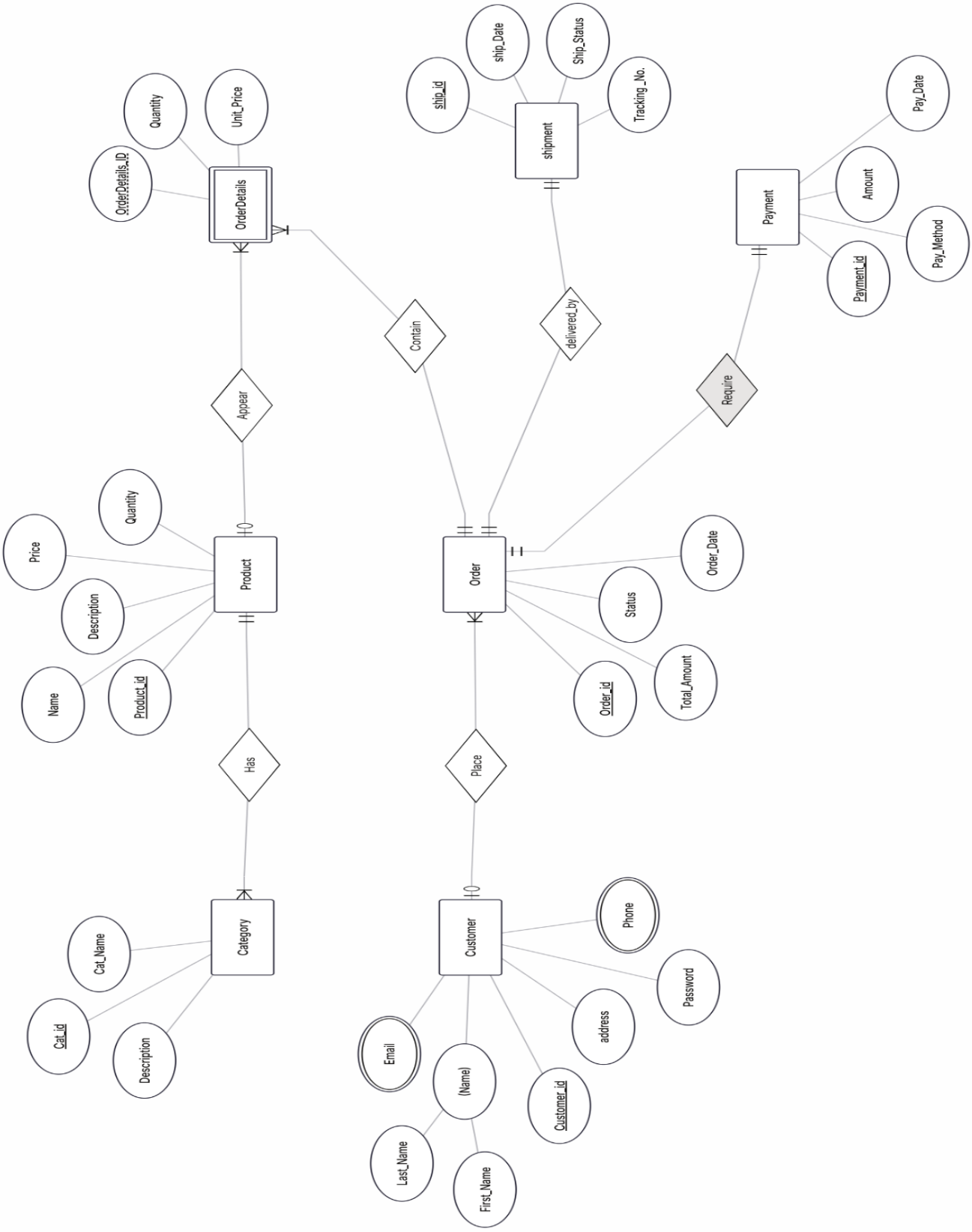
Project Name: Online Store System

Name: Menna Muhammed Ali Eissa

ID: 512395135

Group: 1 (**Monday:** 2:00)

ERD


Quantity
Unit_Price
OrderDetails_ID
OrderDetails
ship_Date
Ship_Status
ship_id
shipment
Tracking_No.
Pay_Date
Amount
Payment
Pay_Method
Payment_id
Contain
delivered_by
Appear
Require
Quantity
Price
Description
Product
Order_Date
Status
Order
Name
Product_id
Order_id
Total_Amount
Has
Place
Cat_Name
Cat_id
Description
Category
Phone
Customer
Password
address
Email
(Name)
Customer_id
Last_Name
First_Name

Mapping

1- Customer

cust_id	Password	First-name	Last-name	Address

Customer phone_number

Cus_id	Cus_number

Customer Email

Cus_id	Email

2- Category

Cat_id	Cat_name	Description

3- Product

Product_id	Name	Description	Price	Quantity	Cat_id

4- OrderDetails

OrderDetails_id	Quantity	Unit-Price	Order_id	Product_id

5- Order

Order_id	Order_date	Total_Amount	Confirm	Cancel	Pending	Customer_id

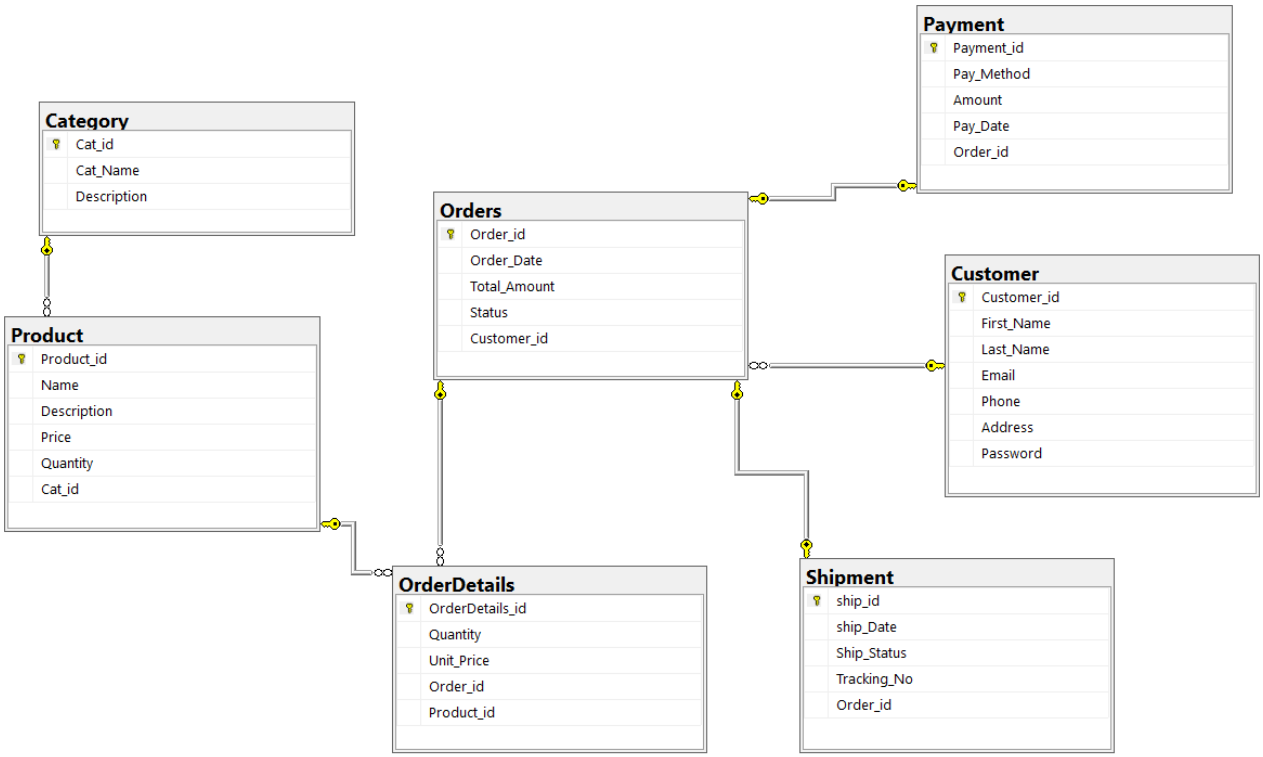
6- Payment

Payment_id	Pay_Method	Pay-Amount	Pay_date	Order_id

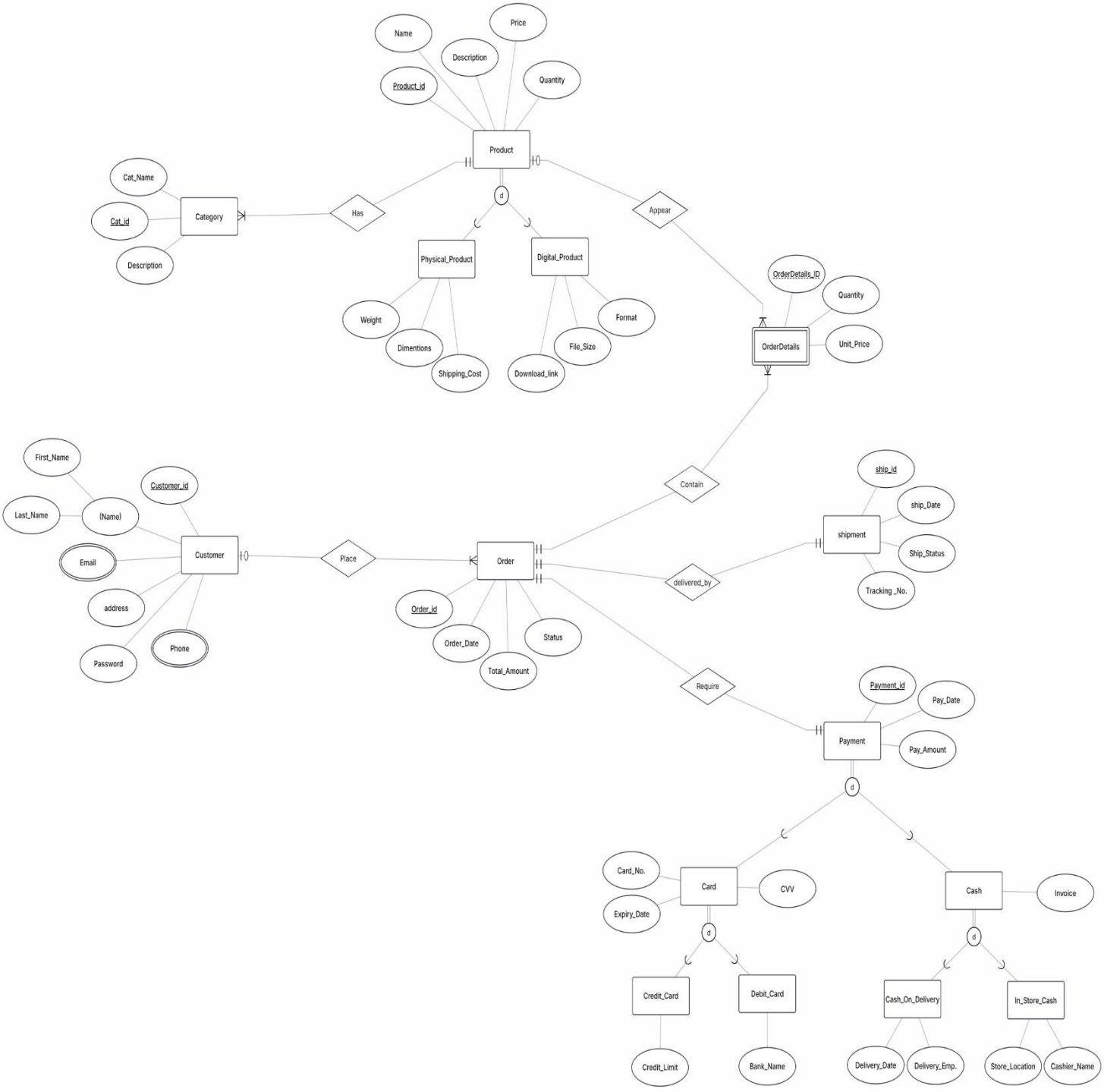
7- Shipment

Ship_id	Ship_data	Status	Tracking_no.	Order_id

SQL


Payment
Payment_id
Pay_Method
Amount
Pay_Date
Order_id
Category
Cat_id
Cat_Name
Description
Orders
Order_id
Order_Date
Total_Amount
Status
Customer_id
Customer
Customer_id
First_Name
Last_Name
Email
Phone
Address
Password
Product
Product_id
Name
Description
Price
Quantity
Cat_id
OrderDetails
OrderDetails_id
Quantity
Unit_Price
Order_id
Product_id
Shipment
ship_id
ship_Date
Ship_Status
Tracking_No
Order_id

EERD


Name
Price
Description
Product_id
Quantity
Product
Cat_Name
Cat_id
Category
Description
Has
d
Appear
Physical_Product
Digital_Product
Weight
Dimentions
Shipping_Cost
Format
File_Size
Download_link
OrderDetails_ID
Quantity
OrderDetails
Unit_Price
First_Name
Customer_id
Last_Name
(Name)
Email
Customer
address
Password
Phone
Place
Order
Order_id
Order_Date
Total_Amount
Status
Contain
delivered_by
ship_id
ship_Date
shipment
Ship_Status
Tracking _No.
Require
Payment_id
Pay_Date
Payment
Pay_Amount
Card_No.
Expiry_Date
Card
CVV
Credit_Card
Debit_Card
Credit_Limit
Bank_Name
Cash
Invoice
Cash_On_Delivery
In_Store_Cash
Delivery_Date
Delivery_Emp.
Store_Location
Cashier_Name

SQL Query

SQLQuery2.s...A\DELL (58))*

```sql
CREATE DATABASE EERD_DB;
USE EERD_DB;

CREATE TABLE Category (
    Cat_id INT PRIMARY KEY,
    Cat_Name VARCHAR(100) NOT NULL,
    Description TEXT
);

CREATE TABLE Product (
    Product_id  INT PRIMARY KEY,
    Name        VARCHAR(150) NOT NULL,
    Description TEXT,
    Price       DECIMAL(10,2) NOT NULL,
    Quantity    INT NOT NULL DEFAULT 0,
    Cat_id      INT,
    FOREIGN KEY (Cat_id) REFERENCES Category(Cat_id)
);

CREATE TABLE Physical_Product (
    Product_id    INT PRIMARY KEY,
    Weight        DECIMAL(8,3),
    Dimensions    VARCHAR(100),
    Shipping_Cost DECIMAL(8,2),
    FOREIGN KEY (Product_id) REFERENCES Product(Product_id)
);
```

SQLQuery2.s...A\DELL (58))*

```sql
CREATE TABLE Digital_Product (
    Product_id    INT PRIMARY KEY,
    File_Size     DECIMAL(10,2),
    Download_Link VARCHAR(500),
    Format        VARCHAR(50),
    FOREIGN KEY (Product_id) REFERENCES Product(Product_id)
);

CREATE TABLE Customer (
    Customer_id INT PRIMARY KEY,
    First_Name  VARCHAR(100) NOT NULL,
    Last_Name   VARCHAR(100) NOT NULL,
    Email       VARCHAR(200) UNIQUE NOT NULL,
    Phone       VARCHAR(20),
    Address     TEXT,
    Password    VARCHAR(255) NOT NULL
);

CREATE TABLE Orders (
    Order_id     INT           PRIMARY KEY,
    Order_Date   DATE          NOT NULL,
    Status       VARCHAR(50)   NOT NULL,
    Total_Amount DECIMAL(12,2) NOT NULL,
    Customer_id  INT           NOT NULL,
    FOREIGN KEY (Customer_id) REFERENCES Customer(Customer_id)
);
```

SQLQuery2.s...A\DELL (58))*

```sql
CREATE TABLE OrderDetails (
    OrderDetails_ID INT           PRIMARY KEY,
    Order_id        INT           NOT NULL,
    Product_id      INT           NOT NULL,
    Quantity        INT           NOT NULL,
    Unit_Price      DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (Order_id)   REFERENCES Orders(Order_id),
    FOREIGN KEY (Product_id) REFERENCES Product(Product_id)
);

CREATE TABLE Shipment (
    ship_id     INT PRIMARY KEY,
    ship_Date   DATE,
    Ship_Status VARCHAR(50),
    Tracking_No VARCHAR(100),
    Order_id    INT         NOT NULL,
    FOREIGN KEY (Order_id) REFERENCES Orders(Order_id)
);

CREATE TABLE Payment (
    Payment_id INT           PRIMARY KEY,
    Pay_Date   DATE          NOT NULL,
    Pay_Amount DECIMAL(12,2) NOT NULL,
    Order_id   INT           NOT NULL,
    FOREIGN KEY (Order_id) REFERENCES Orders(Order_id)
);
```

SQLQuery2.s...A\DELL (58))*

```sql
CREATE TABLE Card (
    Payment_id  INT         PRIMARY KEY,
    Card_No     VARCHAR(20) NOT NULL,
    Expiry_Date DATE        NOT NULL,
    CVV         VARCHAR(5)  NOT NULL,
    FOREIGN KEY (Payment_id) REFERENCES Payment(Payment_id)
);

CREATE TABLE Credit_Card (
    Payment_id   INT           PRIMARY KEY,
    Credit_Limit DECIMAL(12,2),
    FOREIGN KEY (Payment_id) REFERENCES Card(Payment_id)
);

CREATE TABLE Debit_Card (
    Payment_id INT          PRIMARY KEY,
    Bank_Name  VARCHAR(100),
    FOREIGN KEY (Payment_id) REFERENCES Card(Payment_id)
);

CREATE TABLE Cash (
    Payment_id INT PRIMARY KEY,
    Invoice    VARCHAR(100),
    FOREIGN KEY (Payment_id) REFERENCES Payment(Payment_id)
);
```

SQLQuery2.s...A\DELL (58))*

```sql
CREATE TABLE Cash_On_Delivery (
    Payment_id    INT          PRIMARY KEY,
    Delivery_Date DATE,
    Delivery_Emp  VARCHAR(100),
    FOREIGN KEY (Payment_id) REFERENCES Cash(Payment_id)
);

CREATE TABLE In_Store_Cash (
    Payment_id     INT          PRIMARY KEY,
    Store_Location VARCHAR(200),
    Cashier_Name   VARCHAR(100),
    FOREIGN KEY (Payment_id) REFERENCES Cash(Payment_id)
);

INSERT INTO Category VALUES
(1, 'Electronics',    'Electronic devices and accessories'),
(2, 'Clothing',       'Men, women and children apparel'),
(3, 'Books',          'Printed and digital books'),
(4, 'Home Furniture', 'Furniture and home decor'),
(5, 'Cosmetics',      'Skin and hair care products');

INSERT INTO Product VALUES
(101, 'Lenovo IdeaPad Laptop',         'Core i5 12th Gen, 8GB RAM, 512GB SSD',          18999.00,  30,   1),
(102, 'Egyptian Cotton Galabiya',      '100% cotton traditional galabiya, all sizes',     250.00,  200,   2),
(103, 'Azazeel Novel by Youssef Ziedan','Arabic Booker Prize winning historical novel',    120.00,  150,   3),
(104, '5-Piece Sofa Set',              'Modern beech wood frame with velvet upholstery', 12500.00,  15,   4),
(105, 'NIVEA Body Cream 200ml',        'Deep moisturizing cream for face and body',        89.00,  500,   5),
(106, 'Programming in Arabic PDF Book','Complete Arabic programming course - PDF file',   199.00, 9999,   3),
(107, 'JBL Tune 510BT Headphones',     'Wireless Bluetooth headphones, high quality',    1350.00,   80,   1);

INSERT INTO Physical_Product VALUES
(101, 2.100, '37 x 25 x 2 cm',   150.00),
(102, 0.400, '35 x 25 x 5 cm',    25.00),
(103, 0.350, '21 x 14 x 2 cm',    20.00),
(104, 85.000,'210 x 95 x 90 cm', 350.00),
(105, 0.220, '10 x 6 x 6 cm',     15.00),
(107, 0.180, '20 x 18 x 4 cm',    30.00);

INSERT INTO Digital_Product VALUES
(106, 45.5, 'https://downloads.example.eg/programming-arabic.pdf', 'PDF');

INSERT INTO Customer VALUES
(1, 'Ahmed',   'El-Sayed',   'ahmed.elsayed@gmail.com',    '01012345678', '15 Tahrir Street, Downtown, Cairo',      '$2b$12$hashedpassword1'),
(2, 'Mona',    'Ibrahim',    'mona.ibrahim@yahoo.com',     '01098765432', '8 Road 9, Maadi, Cairo',                 '$2b$12$hashedpassword2'),
(3, 'Karim',   'Abdallah',   'karim.abdallah@outlook.com', '01155556789', '42 Haram Street, Giza',                  '$2b$12$hashedpassword3'),
(4, 'Sara',    'Mahmoud',    'sara.mahmoud@hotmail.com',   '01234567890', '27 Abbas El-Akkad, Nasr City, Cairo',    '$2b$12$hashedpassword4'),
(5, 'Mohamed', 'Othman',     'mohamed.othman@gmail.com',   '01501112233', '3 Corniche Road, Sidi Bishr, Alexandria','$2b$12$hashedpassword5');

INSERT INTO Orders VALUES
(1001, '2024-11-01', 'Completed',   18999.00, 1),
(1002, '2024-11-05', 'Out for Delivery', 370.00, 2),
(1003, '2024-11-10', 'Completed',     199.00,  3),
(1004, '2024-11-15', 'Cancelled',   12500.00,  4),
(1005, '2024-11-20', 'Processing',   1439.00,  5);
```

```sql
INSERT INTO OrderDetails VALUES
(5001, 1001, 101, 1, 18999.00),
(5002, 1002, 102, 2,   250.00),
(5003, 1002, 105, 1,    89.00), -- NIVEA bundled with Galabiya order
(5004, 1003, 106, 1,   199.00),
(5005, 1004, 104, 1, 12500.00),
(5006, 1005, 107, 1,  1350.00),
(5007, 1005, 105, 1,    89.00);

INSERT INTO Shipment VALUES
(7001, '2024-11-03', 'Delivered',    'EG-SHIP-00291', 1001),
(7002, '2024-11-07', 'In Transit',   'EG-SHIP-00302', 1002),
(7003, '2024-11-11', 'Delivered',    'EG-SHIP-00318', 1003),
(7004, '2024-11-21', 'Dispatched',   'EG-SHIP-00335', 1005);

INSERT INTO Payment VALUES
(9001, '2024-11-01', 18999.00, 1001),
(9002, '2024-11-05',   370.00, 1002),
(9003, '2024-11-10',   199.00, 1003),
(9004, '2024-11-20',  1439.00, 1005);

INSERT INTO Card VALUES
(9001, '4111111111111111', '2026-08-01', '123'),  -- Ahmed  - Visa credit card
(9003, '5500005555555559', '2025-12-01', '456');  -- Karim  - Mastercard debit

INSERT INTO Credit_Card VALUES
(9001, 50000.00);  -- Ahmed's credit card limit (EGP)
```

SQLQuery2.s...A\DELL (58))*

```sql
(9003, 'National Bank of Egypt');  -- Karim's debit card

INSERT INTO Cash VALUES
(9002, 'INV-EG-2024-1002'),  -- Mona  - cash on delivery
(9004, 'INV-EG-2024-1005');  -- Mohamed - in-store cash

INSERT INTO Cash_On_Delivery VALUES
(9002, '2024-11-07', 'Mostafa Ali');  -- Delivery employee

INSERT INTO In_Store_Cash VALUES
(9004, 'Mall of Arabia Branch, Giza', 'Norhan Hassan');

SELECT * FROM Category;
SELECT * FROM Product;
Select * FROM Physical_Product;
SELECT * FROM Digital_Product;
SELECT * FROM Customer;
SELECT * FROM Orders;
SELECT * FROM OrderDetails;
SELECT * FROM Shipment;
SELECT * FROM Payment;
SELECT * FROM Card;
SELECT * FROM Credit_Card;
SELECT * FROM Debit_Card;
SELECT * FROM Cash;
SELECT * FROM Cash_On_Delivery;
SELECT * FROM In_Store_Cash;
```

Tables

Results Messages

	Cat_id	Cat_Name	Description
1	1	Electronics	Electronic devices and accessories
2	2	Clothing	Men, women and children apparel
3	3	Books	Printed and digital books
4	4	Home Furniture	Furniture and home decor
5	5	Cosmetics	Skin and hair care products

	Product_id	Name	Description	Price	Quantity	Cat_id
1	101	Lenovo IdeaPad Laptop	Core i5 12th Gen, 8GB RAM, 512GB SSD	18999.00	30	1
2	102	Egyptian Cotton Galab...	100% cotton traditional galabiya, all sizes	250.00	200	2
3	103	Azazeel Novel by Yous...	Arabic Booker Prize winning historical n...	120.00	150	3
4	104	5-Piece Sofa Set	Modern beech wood frame with velvet u...	12500.00	15	4
5	105	NIVEA Body Cream 20...	Deep moisturizing cream for face and b...	89.00	500	5
6	106	Programming in Arabi...	Complete Arabic programming course ...	199.00	9999	3
7	107	JBL Tune 510BT Hea...	Wireless Bluetooth headphones, high q...	1350.00	80	1

	Product_id	Weight	Dimensions	Shipping_Cost
1	101	2.100	37 x 25 x 2 cm	150.00
2	102	0.400	35 x 25 x 5 cm	25.00
3	103	0.350	21 x 14 x 2 cm	20.00
4	104	85.000	210 x 95 x 9...	350.00
5	105	0.220	10 x 6 x 6 cm	15.00
6	107	0.180	20 x 18 x 4 cm	30.00

	Product_id	File_Size	Download_Link	Format
1	106	45.50	https://downloads.example.eg/programming-arabic....	PDF

	Customer_id	First_Name	Last_Name	Email	Phone	Address	Password
1	1	Ahmed	El-Sayed	ahmed.elsayed@gmail.com	01012345678	15 Tahrir Street, Downtown, Cairo	$2b$12$hashedpassword1
2	2	Mona	Ibrahim	mona.ibrahim@yahoo.com	01098765432	8 Road 9, Maadi, Cairo	$2b$12$hashedpassword2
3	3	Karim	Abdallah	karim.abdallah@outlook.c...	01155556789	42 Haram Street, Giza	$2b$12$hashedpassword3
4	4	Sara	Mahmoud	sara.mahmoud@hotmail.c...	01234567890	27 Abbas El-Akkad, Nasr City, C...	$2b$12$hashedpassword4
5	5	Mohamed	Othman	mohamed.othman@gmail...	01501112233	3 Corniche Road, Sidi Bishr, Ale...	$2b$12$hashedpassword5

	Order_id	Order_Date	Status	Total_Amount	Customer_id
1	1001	2024-11-01	Completed	18999.00	1
2	1002	2024-11-05	Out for Delivery	370.00	2
3	1003	2024-11-10	Completed	199.00	3
4	1004	2024-11-15	Cancelled	12500.00	4
5	1005	2024-11-20	Processing	1439.00	5

	OrderDetails_ID	Order_id	Product_id	Quantity	Unit_Price
1	5001	1001	101	1	18999.00
2	5002	1002	102	2	250.00
3	5003	1002	105	1	89.00
4	5004	1003	106	1	199.00
5	5005	1004	104	1	12500.00
6	5006	1005	107	1	1350.00
7	5007	1005	105	1	89.00

	ship_id	ship_Date	Ship_Status	Tracking_No	Order_id
1	7001	2024-11-03	Delivered	EG-SHIP-00291	1001
2	7002	2024-11-07	In Transit	EG-SHIP-00302	1002
3	7003	2024-11-11	Delivered	EG-SHIP-00318	1003
4	7004	2024-11-21	Dispatched	EG-SHIP-00335	1005

	Payment_id	Pay_Date	Pay_Amount	Order_id
1	9001	2024-11-01	18999.00	1001
2	9002	2024-11-05	370.00	1002
3	9003	2024-11-10	199.00	1003
4	9004	2024-11-20	1439.00	1005

	Payment_id	Card_No	Expiry_Date	CVV
1	9001	4111111111111111	2026-08-01	123
2	9003	5500005555555559	2025-12-01	456

	Payment_id	Credit_Limit
1	9001	50000.00

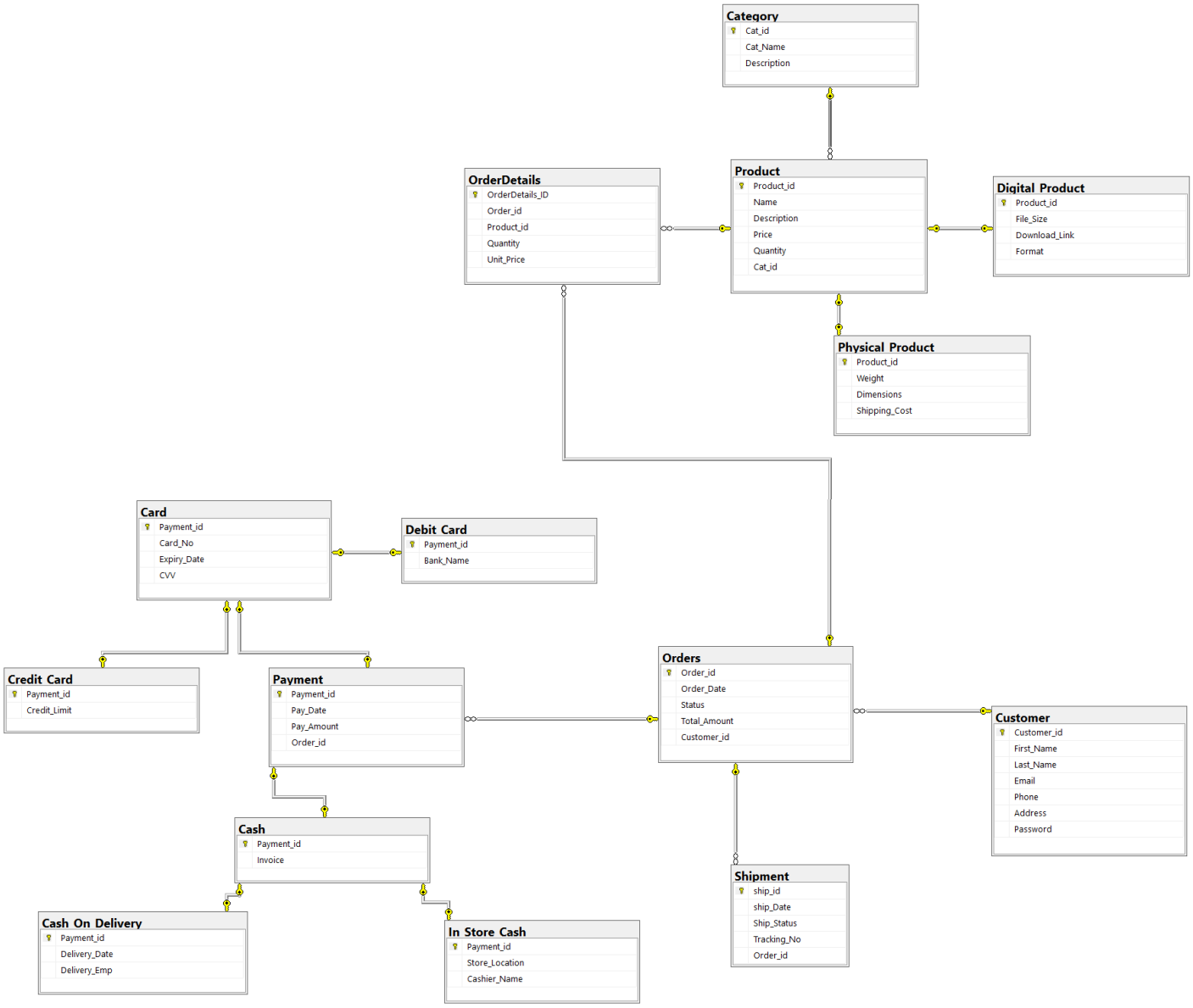
	Payment_id	Bank_Name
1	9003	National Bank of Egypt

	Payment_id	Invoice
1	9002	INV-EG-2024-1002
2	9004	INV-EG-2024-1005

	Payment_id	Delivery_Date	Delivery_Emp
1	9002	2024-11-07	Mostafa Ali

	Payment_id	Store_Location	Cashier_Name
1	9004	Mall of Arabia Branch, Giza	Norhan Hassan

Schema


Category
Cat_id
Cat_Name
Description
OrderDetails
OrderDetails_ID
Order_id
Product_id
Quantity
Unit_Price
Product
Product_id
Name
Description
Price
Quantity
Cat_id
Digital Product
Product_id
File_Size
Download_Link
Format
Physical Product
Product_id
Weight
Dimensions
Shipping_Cost
Card
Payment_id
Card_No
Expiry_Date
CVV
Debit Card
Payment_id
Bank_Name
Credit Card
Payment_id
Credit_Limit
Payment
Payment_id
Pay_Date
Pay_Amount
Order_id
Orders
Order_id
Order_Date
Status
Total_Amount
Customer_id
Customer
Customer_id
First_Name
Last_Name
Email
Phone
Address
Password
Cash
Payment_id
Invoice
Shipment
ship_id
ship_Date
Ship_Status
Tracking_No
Order_id
Cash On Delivery
Payment_id
Delivery_Date
Delivery_Emp
In Store Cash
Payment_id
Store_Location
Cashier_Name